

Worksheet: Knowledge Representation & Ontologies

Foundations of Artificial Intelligence – SS 2026

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Instructions: Written tasks (✎) answer here or in a Markdown cell. Programming tasks (🐍) go in `ontology_exercises.ipynb`. Estimated time: ~45 minutes.

Part 1 – Conceptual Questions ✎ (~10 min)

1.1 What are the three components of a Description Logic knowledge base? Briefly describe what each stores.

1.2 Consider the following facts:

```
Ben is a human.      Sarah is a human.
Ben owns Rex.        Sarah owns Kittypillar.
Rex is a dog.        Kittypillar is a cat.
```

Answer these questions under **both** assumptions and explain the difference:

Question	Closed-World Assumption	Open-World Assumption
Does Ben own a dog?		
Does Ben own a cat?		
Is Rex a cat?		

1.3 What is the difference between an **Ontology** and a **Knowledge Graph**? Use the TBox/ABox terminology in your answer.

1.4 Match each ALC construct to its meaning:

Construct	Meaning
$\top$	
$\bot$	
$C \sqcap D$	
$C \sqcup D$	
$\exists r.C$	
$\forall r.C$	
$C \sqsubseteq D$	

1.5 Identify whether each of the following belongs to the **TBox**, **ABox**, or **RBox**:

Statement	T / A / R
<code>Cat $\sqsubseteq$ Mammal</code>	
<code>Mother(julia)</code>	
<code>dreams $\sqsubseteq$ sleeps</code>	
<code>HappyCatOwner $\sqsubseteq$ $\exists$owns.Cat</code>	
<code>ParentOf(julia, max)</code>	
<code>Dis(sleeps, runs)</code>	

Part 2 – Reading DL Axioms (~5 min)

Translate each DL axiom into plain English:

- (a) `Lecturer  $\sqsubseteq$   $\exists$ teaches.Lecture`
- (b) `Mother  $\equiv$  Female  $\sqcap$  Parent`
- (c) `Male  $\sqcap$  Female  $\sqsubseteq \perp$`
- (d) `HappyCatOwner  $\sqsubseteq$   $\exists$ owns.Cat  $\sqcap$   $\forall$ caresFor.Healthy`
- (e) `owns  $\sqsubseteq$  caresFor` (*RBox*)

Part 3 – Writing DL Axioms (~5 min)

Write DL axioms (TBox / ABox / RBox entries) for the following statements. Use your own concept and role names.

- (a) Every student is a person.
- (b) A vegetarian is a person who only eats plant-based food.
- (c) No person can be both alive and dead at the same time.
- (d) Every robot has at least one sensor.
- (e) If a robot perceives something, it is aware of it. (*RBox*)
- (f) The individual `tiago` is a robot located in the kitchen. (*ABox*)

Part 4 – Design Your Own Ontology (~25 min)

Open `ontology_exercises.ipynb` and complete **Exercise 4**.

You will build a small **kitchen robot ontology** using Python's `owlready2` library, then run a reasoner to derive new facts automatically.

4.1 Define the TBox — add the classes and properties provided, then add **two of your own**.

4.2 Populate the ABox — add the individuals provided, then add **one of your own**.

4.3 Run the HermiT reasoner and inspect what new facts were inferred.

4.4 (Bonus): Introduce an inconsistency (e.g. make an individual both `Alive` and `Dead`) and observe what the reasoner reports.

Good luck! 🎓